

CONNECTING THE DOTS

*A Knowledge Graph for Exposing Fraud-Linked
Transaction Patterns in Banking Data*

Victoria Bassey • 948253

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5M+

TRANSACTIONS

5,177

FRAUD CASES

\$186B

LAUNDERED

243×

TOP SUSPECT



Why Flat Data Hides Fraud

The Flat CSV Problem

Fraudulent transactions are structurally designed to appear as noise — isolated, normal-looking rows in a relational database. A single transaction reveals nothing. A network of semantically linked transactions reveals the pattern.

The problem is not the data. It is the shape of the data.

Flat CSV forces investigators to look at rows. Knowledge graphs allow investigators to see networks. Fraud lives in the network, not the rows.

THE SOLUTION



Raw CSV (5M rows)



Python → RDF Triples



GraphDB Triplestore



SPARQL Queries



Fraud Patterns Exposed

CORE ARGUMENT: Flat CSV represents rows. Knowledge graphs represent relationships. Semantic structures expose what isolated records conceal.

IBM Anti-Money Laundering Dataset

Synthetic dataset by IBM Research (Kaggle). Designed for AML research with explicit laundering labels. Covers 7 payment formats across real-world transaction patterns.

5,078,345

Total Transactions

5,177

Labelled Laundering

10,177

In KG

7

Payment Formats

♦ Stratified sampling: All 5,177 laundering cases retained + 5,000 random normal transactions. No fraud signal excluded.

DATA QUALITY CHECK

Missing Values

0

Negative Amounts

0

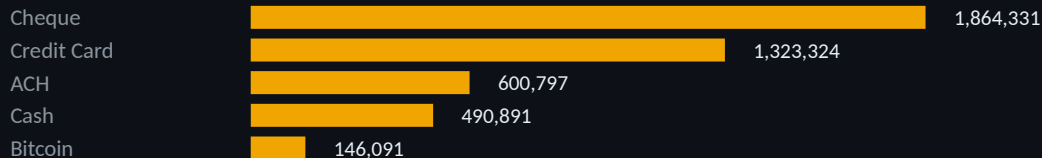
Duplicates Removed

9

Max Single Transaction

\$1T+

Payment Format Distribution



OWL Ontology in Protégé

Ontology IRI: <http://www.semanticweb.org/banking-fraud-kg#>

Bank

Account

Transaction

PaymentFormat

Currency

FraudIndicator

LaunderingPattern

Entity

Key Design Decisions

- ✓ Domain & range restrictions on all object properties
- ✓ Disjoint class axioms — prevents wrong inferences
- ✓ FIBO used as reference vocabulary
- ✓ HermiT reasoner result: STABLE — no inconsistencies



HermiT Reasoner

STABLE

No inconsistencies • No spurious inferences • Fully validated

How Money Launderers Hide the Trail

FAN OUT



One account sends money to several different accounts at the same time — splitting the funds to make each transfer look unrelated.

WHAT HAPPENED IN THE DATA

Acct_800737690	→	Acct_80011F990	\$2,848
Acct_800737690	→	Acct_80020C5B0	\$4,200
Acct_800737690	→	Acct_8000EBD30	\$5,100

Same sender · 3 destinations · within 24 hrs

CYCLE



Money moves through a chain of accounts and ends up back where it started — making it nearly impossible to trace where the funds originally came from.

WHAT HAPPENED IN THE DATA

Acct_80020C5B0	→	Acct_800737690	\$8,000
Acct_800737690	→	Acct_8000EBD30	\$7,900
Acct_80098A3D0	→	Acct_800737690	\$9,100

Amounts vary each time to avoid triggering alerts

GATHER SCATTER



Money is collected from many different accounts into one, then sent back out to many others — so no single payer can be linked to any single receiver.

WHAT HAPPENED IN THE DATA

Acct_800EE6B70	→ FUNNEL	\$7,600
Acct_8000EBD30	→ FUNNEL	\$6,400
FUNNEL →	Acct_80098A3D0	\$6,900
Acct_800EE6B70	→ FUNNEL	\$18,721
FUNNEL →	Acct_80098A3D0	\$18,030

Funnel = Acct_80044A5D0

Q1: Full Network — All 12 Transactions at a Glance

What this query does

Retrieves every transaction entity in the prototype graph with its linked sender and receiver accounts — rendering the complete semantic network in a single query. In flat CSV, these 12 records are disconnected rows. In the RDF knowledge graph, the linked entity structure surfaces three co-existing laundering patterns simultaneously.

KEY INSIGHT

Rows 1, 4 & 5: Account_800737690 is semantically linked to three distinct receiver entities — the FanOut structure is immediately visible through graph traversal. Rows 8, 9 & 12 form the Cycle. Rows 2, 3, 6, 7, 11 constitute the GatherScatter subgraph.

GRAPHDB RESULT

Transaction	Sender Account	Receiver Account
Transaction_001	Account_800737690	Account_80011F990
Transaction_002	Account_800EE6B70	Account_80044A5D0
Transaction_003	Account_80044A5D0	Account_80098A3D0
Transaction_004	Account_800737690	Account_80020C5B0
Transaction_005	Account_800737690	Account_8000EBD30
Transaction_006	Account_800EE6B70	Account_80044A5D0
Transaction_007	Account_80044A5D0	Account_80098A3D0
Transaction_008	Account_80020C5B0	Account_800737690
Transaction_009	Account_800737690	Account_8000EBD30
Transaction_010	Account_80011F990	Account_80020C5B0
Transaction_011	Account_8000EBD30	Account_80044A5D0
Transaction_012	Account_80098A3D0	Account_800737690

Q2: FanOut — One Sender, Multiple Receivers

What this query does

Filters only transactions labelled Pattern_FanOut — grouping by sender to reveal accounts dispersing money in multiple directions simultaneously. No manual searching. The SPARQL surfaces the suspect automatically.

KEY INSIGHT

Account_800737690 is semantically linked to three distinct receiver accounts within a 24-hour window — \$2,848.96, \$4,200.00, and \$5,100.00. This one-to-many transaction structure is consistent with fund dispersal patterns associated with placement-stage laundering.

GRAPHDB RESULT

Sender	#	Receiver	Amount	Timestamp
Acct_80011F990	1	Acct_80020C5B0	\$3,200.00	2022-09-05 10:12
Acct_800737690	1	Acct_80011F990	\$2,848.96	2022-09-01 00:06
Acct_800737690	1	Acct_80020C5B0	\$4,200.00	2022-09-02 09:15
Acct_800737690	1	Acct_8000EBD30	\$5,100.00	2022-09-02 11:20



$\$2,848 \cdot \$4,200 \cdot \$5,100$ — three directions, one linked entity

Q3: Cycle — Money Looping Through Accounts

What this query does

Retrieves all transactions tagged Pattern_Cycle, ordered by timestamp. This exposes the circular flow of funds — money that leaves an account, passes through intermediaries, and eventually returns to its origin to be used again.

KEY INSIGHT

The transaction amounts vary across each leg: \$8,000 → \$7,900 → \$9,100. This variation is structurally significant — identical repeated amounts are a common flag in rule-based AML systems. The graph traversal exposes the circular flow regardless of amount variation.

GRAPHDB RESULT

Transaction	Sender	Receiver	Amount	Timestamp
Txn_008	Acct_80020C5B0	Acct_800737690	\$8,000.00	2022-09-04 09:25
Txn_009	Acct_800737690	Acct_8000EBD30	\$7,900.00	2022-09-04 13:40
Txn_012	Acct_80098A3D0	Acct_800737690	\$9,100.00	2022-09-06 11:32



 \$9,100 returns via Acct_80098A3D0 (Sep 6)

Q4: GatherScatter

Q5: Fraud Leaderboard

Q4 GatherScatter — Funnel Account Detection

Account_80044A5D0 functions as a semantic hub: it holds incoming edges from multiple sender entities and outgoing edges to multiple receiver entities simultaneously — severing any direct source-to-destination provenance in the graph.

Transaction	Sender	Receiver	Amount	Date
Txn_006	Acct_800E E6B70	Acct_8004 4A5D0	\$7,600.00	Sep-03 08:45
Txn_007	Acct_8004 4A5D0	Acct_8009 8A3D0	\$6,900.00	Sep-03 14:10
Txn_011	Acct_8000 EBD30	Acct_8004 4A5D0	\$6,400.00	Sep-05 16:18
Txn_002	Acct_800E E6B70	Acct_8004 4A5D0	\$18,721.7 8	Sep-06 08:41
Txn_003	Acct_8004 4A5D0	Acct_8009 8A3D0	\$18,030.2 5	Sep-07 16:48

Money flows IN from many · flows OUT to many · no single source traces to any destination.

Q5 Most Suspicious Accounts — Fraud Leaderboard

Counts every account's appearances as sender OR receiver across all confirmed laundering transactions — auto-ranking each account by suspicion level.

Account	Times Appeared
Account_800737690	6
Account_80044A5D0	5
Account_80020C5B0	3
Account_8000EBD30	3
Account_80098A3D0	3
Account_800EE6B70	2
Account_80011F990	2

Account_800737690 appears 6 times across 12 transactions — as both sender and receiver, and across two different pattern types. An account this deeply embedded in multiple overlapping laundering structures is the clearest signal the graph can surface.

Q6: Pattern Summary — Financial Footprint of the Operation

What this query does

Aggregates all transaction entities by their associated laundering pattern class — counting members and summing amounts per pattern. This produces a semantic summary of the entire suspicious subgraph, including transaction volume and financial weight per pattern type.

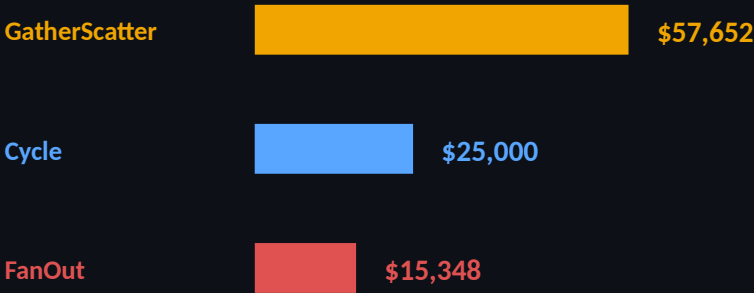
KEY INSIGHT

GatherScatter accounts for the highest total value (\$57,652) with the fewest transactions — indicating high-value fund consolidation and redistribution. Total: \$98,000 in semantically flagged suspicious movement across 12 prototype transactions, surfaced entirely through SPARQL queries on the linked graph.

GRAPHDB RESULT

Pattern	# Transactions	Total Laundered
Pattern_GatherScatter	5	\$57,652.03
Pattern_Cycle	3	\$25,000.00
Pattern_FanOut	4	\$15,348.96

Money moved per pattern:



TOTAL: \$98,000.99 suspicious movement • 12 transactions • zero manual analysis

12 Transactions. One Suspect. Three Fraud Stories.

FAN OUT

Acct_800737690 → Acct_80011F990	\$2,848.96
Acct_800737690 → Acct_80020C5B0	\$4,200.00
Acct_800737690 → Acct_8000EBD30	\$5,100.00

All sent within a 24-hour window. One source, three destinations.

Semantically: one entity linked outward to three separate receiver nodes.



Account_800737690

PRIMARY SUSPECT

CYCLE

Acct_80020C5B0 to Acct_800737690	\$8,000.00
Acct_800737690 to Acct_8000EBD30	\$7,900.00
Acct_80098A3D0 to Acct_800737690	\$9,100.00

Money returns to origin. Amounts vary each leg to avoid triggering rule-based alert thresholds.

Semantically: a directed cycle — each account is both sender and receiver.

6×

appearances in
12 transactions

\$98K

suspicious money
across 12 transactions

BOTH

FanOut + Cycle
patterns active



Confirmed across
full 5,177 dataset

Scaling to 5,177 Laundering Cases

FRAUD LEADERBOARD — TOP 5 ACCOUNTS

#1	Account_100428660	
#2	Account_1004286A8	
#3	Account_811C599A0	52
#4	Account_80266F880	51
#5	Account_8021353D0	49

Account_100428660 — 243 appearances across confirmed laundering transactions, surfaced via a single SPARQL query traversing the linked graph. The same analysis on flat CSV would require sequential row-by-row scanning.

LAUNDERED BY PAYMENT FORMAT

Payment Format	Transactions	Total Laundered	Share
ACH	4,491	\$186.6B	87%
Cheque	324	\$178.5M	6%
Cash	108	\$255.4M	<2%
Credit Card	206	\$5.0M	<4%
Wire	4	\$31.5K	<1%
Bitcoin	56	\$35.02	<1%

87%

of all laundering used ACH. Fast, widely available, harder to trace than cash. \$186 BILLION through one payment channel.

Hub Accounts & Mega-Transactions

HUB ACCOUNTS — Sending AND Receiving

Account_80266F880

Sent: 29

Recv: 22

● ACTIVE HUB

Account_811B83280

Sent: 17

Recv: 17

● BALANCED CYCLING

Account_811C59700

Sent: 23

Recv: 23

● BALANCED CYCLING

Account_800737690

Sent: 20

Recv: 2

● ACTIVE HUB

Account_800737690 — identified as high-degree in the prototype graph and confirmed as such in the full 5,177-transaction dataset. The semantic structure was consistent across both scales.

LARGEST SINGLE LAUNDERING TRANSACTIONS

Txn_000236

\$84.8B

Txn_000442

\$66.1B

Txn_002598

\$21.5B

Txn_003025

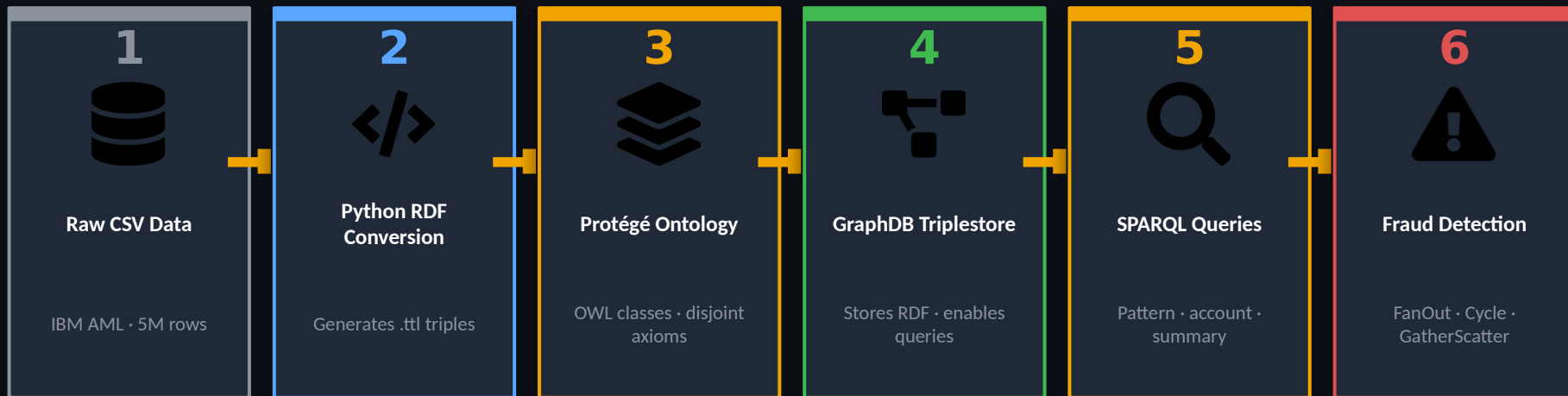
\$2.6B

Txn_001512

\$2.5B

\$84.8 billion. One transaction. One row among 5 million. SPARQL graph traversal surfaced it immediately within the linked structure — without any manual scanning.

The Complete Pipeline



Raw → Fragmented

Flat CSV: disconnected rows, no relationships visible

Semantic → Connected

RDF triples link every entity: accounts, banks, transactions

Query → Explainable

SPARQL traverses the linked graph structure — every result is transparent and traceable back to the ontology

Constraints and Scope Boundaries

Every academic project has boundaries. These are ours — what this prototype does well, where it stops, and what a real-world version would need to go further.

L1**Synthetic Dataset**

The data used is artificially created and pre-labelled by IBM. In the real world, fraudulent transactions are not labelled — they have to be identified without knowing in advance which ones are suspicious.

L2**Prototype-Scale Ontology**

The ontology was built for this prototype with 8 classes. A real banking system would need a much larger, more detailed structure covering hundreds of entity types, legal relationships, and jurisdiction-specific rules.

L3**No Temporal Reasoning**

The graph stores timestamps but cannot reason about time. For example, it cannot automatically detect that a pattern escalated over three weeks, or that transactions happened suspiciously close together across accounts.

L4**Stratified Sampling for Scalability**

Only 10,177 of over 5 million transactions were loaded into the graph. Loading everything would require much more powerful infrastructure than a prototype setup, and queries would need to be optimised for that scale.

L5**No Real Compliance Integration**

This project is a research prototype. It has not been connected to any real banking system or legal compliance framework. The patterns it identifies are for academic demonstration — not for use in actual fraud investigations.

What the Knowledge Graph Proved

Objective 1 Achieved



SPARQL queries traversed the semantic graph and surfaced FanOut, Cycle, and GatherScatter structures — invisible in flat relational data. Account_800737690 was identified as a high-degree linked entity in the prototype and confirmed in the full dataset.

Objective 2 Achieved



Account_100428660 was linked to 243 confirmed laundering transactions through SPARQL graph traversal across 5,177 transaction entities — a query that would require sequential manual scanning in a flat relational structure.

Objective 3 Achieved



The HermiT OWL reasoner confirmed zero ontological inconsistencies and produced no spurious inferences. The graph is semantically consistent, and every query result is explainable and traceable through the ontology structure.

The knowledge graph provides transparent, explainable, and traceable relationship analysis — without relying on opaque predictive models.

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